

MACHINE LEARNING FOR SOCIAL MEDIA EMOTION ANALYSIS

Tracking Public Emotional Distress

During the 2026 Iran-Israel-US War via NLP Analysis of Reddit Discourse

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Abstract

This study presents a comparative NLP framework spanning five methods to quantify mental-health signals in Reddit discourse during the 2026 Iran–Israel–US war. From 208,386 raw comments, a structured cleaning and modelling pipeline produced a fine-tuned DistilBERT classifier achieving a test F1 of 0.9332 on the Saravia emotion benchmark, outperforming BERT (0.9282) at approximately half the training cost. Applied to 96,879 war-period comments and benchmarked against a 3,083-comment pre-war baseline (April 15 to June 10, 2025), the model estimates a +4.2 percentage point increase in distress ratio (chi-square $p = 1.72e-06$). However, cross-model validation using an independent neutral-aware HuggingFace model (j-hartmann; 7 classes) reduces this shift to +0.9pp ($p = 0.069$, not statistically significant), indicating that the fine-tuned model overestimates distress due to forced-emotional-label bias. While the analysis uses the fine-tuned model, conclusions are based on the cross-validated estimate, indicating a modest and statistically weak increase in distress.

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GITHUB REPOSITORY LINK: <https://github.com/apurwakumari16nus/ML-Assignment2>

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1. Introduction and Research Context

Geopolitical crises unfold in near-real-time on social media, yet traditional methods for measuring their emotional toll on populations such as surveys, clinical data, and after-action reports, are slow, expensive, and lag behind events by weeks or months. Natural language processing of public social-media text offers a scalable, near-real-time alternative for monitoring population-level emotional well-being during crises.

This project develops an end-to-end NLP pipeline to measure emotional shifts on Reddit during the 2026 Iran-Israel-US war, while also stress-testing its conclusions through independent cross-model validation, a step that materially deflates the headline finding and represents the key methodological contribution of this work. The pipeline spans data collection, cleaning, model training, inference, temporal and cross-community analysis, event-level significance testing, and neutral-aware cross-model validation.

This work extends the methodology of Wang et al. (2025, Nature Scientific Reports), who used BERT to track emotional responses during the 2023 Israel-Gaza conflict. The contributions are threefold: (1) a five-method comparison spanning a lexicon baseline, two classical machine learning models, and two Transformer models; (2) a true pre-war baseline anchored in a known non-conflict period, rather than noisy internal variation; and (3) independent cross-model validation using a neutral-aware emotion model to quantify and correct the forced-emotional-label bias inherent in 6-class schemes.

1.1 Research Questions

- **RQ1 (Methodological).** Which method provides the best accuracy-cost trade-off for large-scale social-media emotion analysis across a lexicon baseline, classical machine learning models, and Transformer models?
- **RQ2 (Baseline shift).** Is the overall level of emotional distress higher during the war period compared to a matched pre-war baseline from the same subreddits and keyword filters?
- **RQ3 (Temporal).** Do emotional distress signals change around major conflict events, and are these changes statistically significant after correcting for multiple testing?
- **RQ4 (Community).** Do emotional distress patterns differ between news subreddits (e.g., r/worldnews, r/geopolitics) and mental-health subreddits (e.g., r/mentalhealth, r/anxiety) during the conflict?

1.2 Key Findings

- **Distress levels show a modest increase during the war period.**
Under the fine-tuned model, distress rises from 60.6% (pre-war baseline) to 64.8% (war period), a +4.2pp increase ($p = 1.72e-06$). However, cross-model validation using a neutral-aware model reduces this to +0.9pp ($p = 0.069$, not significant), indicating a weak positive effect that is not statistically robust.
- **DistilBERT provides the best accuracy–cost trade-off.**
Among the five methods evaluated, DistilBERT achieves the highest F1 score while training in roughly half the time of BERT, making it the most efficient model for this task.

- **Event-level effects are not statistically robust.**

While 3 of 8 conflict events show significant changes in distress at $\alpha = 0.05$, none remain significant after Bonferroni correction, suggesting that distress responses are not strongly driven by individual events.

- **News subreddits show slightly higher distress than mental-health subreddits.**

News communities exhibit approximately 2pp higher distress levels, although this comparison should be interpreted cautiously due to smaller sample sizes in mental-health subreddits.

- **A domain mismatch leads to overestimation of distress.**

More than half of Reddit comments are classified as neutral by an independent model, a category absent in the 6-class setup. This results in systematic overestimation of emotional intensity in the fine-tuned model, highlighting the importance of cross-model validation.

1.3 Why F1 is the Primary Classification Metric

The Saravia dataset is imbalanced, with some emotions more common than others. Accuracy alone can therefore be misleading, as a model may favour majority classes. The weighted F1-score accounts for this by balancing precision and recall across classes, so we use F1 as the primary metric and report accuracy for completeness.

Metric note. All reported F1 scores are weighted F1 on the 2,000-row Saravia test split, computed as the mean over three random seeds (42, 123, 456) to control for training-run variance.

2. Study Context and Data

Three data sources are used: (a) Reddit posts and comments from the 2026 war period, (b) a pre-war baseline from the same subreddits, and (c) the dair-ai/emotion (Saravia) corpus for supervised training. Seven subreddits are selected to represent two discourse modes: news/political (r/worldnews, r/geopolitics, r/iran, r/israel, r/middleeast) and mental-health (r/mentalhealth, r/anxiety), enabling the community-level analysis (RQ2).

The pre-war baseline window (April 15 – June 10, 2025) is chosen to meet three criteria: (1) it precedes the June 13, 2025 Israeli strike, ensuring a non-conflict period; (2) it uses identical subreddits and keyword filters as the war-period data to control for composition; and (3) it falls within Reddit's ~12-month API window, ensuring reliable data retrieval.

Source	Window	Posts	Comments (raw)	Role
Reddit (war period)	Feb 1 – Apr 3, 2026	4,441	208,386	Target of analysis
Reddit (peaceful baseline)	Apr 15 – Jun 10, 2025	141	3,740	Pre-war emotional anchor
dair-ai/emotion (Saravia)	2018 (static corpus)	—	16,000 tweets	Supervised training data

Table 1. Data source summary before cleaning. Saravia covers six emotions (sadness, joy, love, anger, fear, surprise) and is used to fit all four 6-class models.

3. Data and Methodology

The analytical framework consists of three stages: deterministic text cleaning, supervised training of five classifiers on the Saravia corpus, and inference on the cleaned Reddit data. All components of the pipeline are open-source, and the text cleaning, training, and inference processes are deterministic given fixed random seeds.

3.1 Cleaning Pipeline

Cleaning is applied in a fixed sequence to both the war-period and baseline Reddit datasets. Each step addresses a specific data-quality issue with a documented rationale. Both datasets pass through the same deterministic cleaning function: deduplication on post_id/comment_id; exclusion of bot accounts (AutoModerator, RemindMeBot, WikiTextBot — $\approx 4\%$ of raw); dropping of [deleted]/[removed]/empty bodies; URL stripping; Markdown flattening; HTML entity decoding; whitespace normalisation; and a 15-character minimum length filter. Symmetric processing ensures any downstream difference reflects a real shift, not a cleaning artefact.

Cleaning Pipeline - War Period Comments

Cleaning Step	Rows Remaining	Change
Raw dataset loaded	208,386	-
After deduplication	103,411	-104,975 (-50.4%)
After post-join filter	103,411	0
After date filter ($\geq$ Feb 2026)	103,411	0
After removing bot authors	99,612	-3,799 (-3.7%)
After body clean + length ≥ 15	96,879	-2,733 (-2.7%)
Final clean dataset	96,879	46.5% of raw data

Table 2. Row counts at each cleaning stage for the war-period comments. Deduplication is the largest reduction, consistent with Reddit search returning overlapping results across paginated keyword queries.

Cleaning Pipeline - Peaceful Baseline Comments

Cleaning Step	Rows Remaining	Change
Raw baseline comments	3,740	-
After dedup + bot removal	3,188	-552
After body clean + length ≥ 15	3,083	-105
Final baseline dataset	3,083	82.4% of raw data

Table 3. Baseline cleaning follows the same rules as the war-period data, ensuring the two samples are processed symmetrically.

3.2 Analytical Framework: Five-Method Comparison

Five NLP methods are implemented across increasing levels of complexity. This comparison addresses RQ1 by evaluating how much accuracy gain Transformers provide on this task, and at what computational cost.

Tier	Method	Parameters / Features	Role
Baseline	VADER lexicon	Rule-based, ~7,500 terms	Sentiment polarity anchor
Classical	TF-IDF + LogReg	32,995 TF-IDF feats, $C = 10.0$	Fast 6-class classifier
Classical	TF-IDF + SVM	32,995 TF-IDF feats, $C = 1.0$	Margin-based 6-class classifier
Transformer	DistilBERT (fine-tuned)	66M parameters, $LR = 5e-5$	Selected Model
Transformer	BERT-base-cased (ft)	110M parameters, $LR = 2e-5$	Higher-capacity benchmark model

Table 4. Method tiers. All 6-class classifiers are trained on the same 16,000-row Saravia training split and evaluated on the same 2,000-row test split.

3.3 Training Protocol and Hyperparameter Tuning

All models are trained and evaluated on the identical split to ensure direct comparability of F1 scores. Reddit data is not used for training and is reserved solely for inference. Neural network training is sensitive to weight initialisation, data shuffling, and dropout. To ensure robustness, each neural model is trained using three random seeds (42, 123, 456), and reported F1 scores are averaged across runs. The resulting standard deviations are low (< 0.002), indicating stable model performance.

Model	Hyperparameter	Grid	Best	Val F1
TF-IDF + LogReg	C (inverse L2)	{0.1, 1.0, 10.0}	10.0	0.8823
TF-IDF + SVM	C (margin)	{0.1, 1.0, 10.0}	1.0	0.8948
BERT	learning rate	{2e-5, 3e-5, 5e-5}	2e-5	0.9278
DistilBERT	learning rate	{2e-5, 3e-5, 5e-5}	5e-5	0.9321

Table 5. Hyperparameter grids and winning values. Classical models tune C; neural models tune learning rate. All tuning is done on the validation split only; the test split is held out for final reporting.

4. Results

4.1 Model Performance

All five methods were evaluated on the Saravia test set. DistilBERT achieved the best overall performance with an F1 score of 0.9332, slightly outperforming BERT (0.9282) while requiring roughly half the training time. Classical models (TF-IDF + Logistic Regression and SVM) performed moderately well, while the VADER lexicon provided a lower baseline. These results establish DistilBERT as the best accuracy–cost trade-off for this task.

Model	Accuracy	F1	Train (s)	Infer (s)	GPU?
★ DistilBERT (fine-tuned)	0.9340	0.9332	319	1.70	Yes
BERT (fine-tuned)	0.9290	0.9282	622	2.22	Yes
TF-IDF + Linear SVM	0.8800	0.8782	0.13	0.0003	No
TF-IDF + LogReg	0.8665	0.8638	1.43	0.0004	No
VADER (3-class sentiment)	0.7555	0.7571	0.00	0.08	No

Table 6. Model performance on the 2,000-comment Saravia test set (F1 averaged over 3 seeds). ★ marks the production model. DistilBERT offers the best accuracy-cost trade-off; VADER serves as a simple polarity baseline.

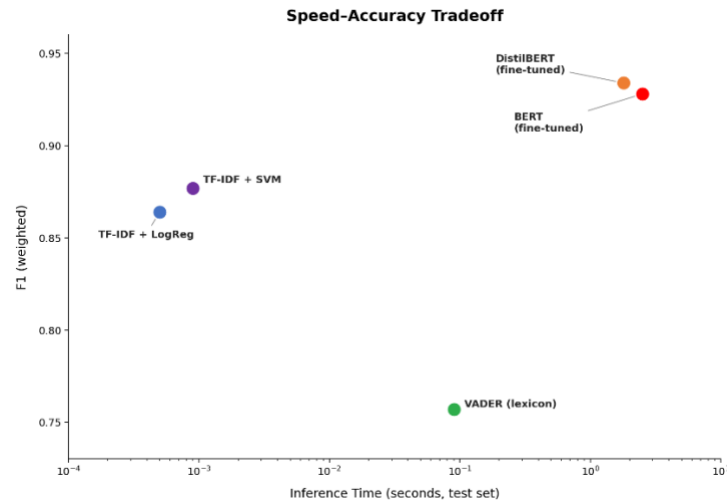


Figure 1. Accuracy-cost tradeoff across all five methods. DistilBERT (★) is the selected model.

4.2 Distress Levels During the War

Applying the fine-tuned DistilBERT model to Reddit data shows a distress ratio of 64.8% during the war period, compared to 60.6% in the pre-war baseline, representing a +4.2 percentage point increase. This difference is statistically significant ($p = 1.72e-06$) under the single-model analysis.

Emotion	Baseline %	War %	Change
sadness	7.2%	8.6%	+1.3pp
joy	37.6%	33.3%	-4.3pp
love	0.7%	0.5%	-0.2pp
anger	47.2%	49.0%	+1.8pp
fear	6.2%	7.3%	+1.1pp

Emotion	Baseline %	War %	Change
surprise	1.0%	1.3%	+0.3pp

Table 7. Change in emotion distribution from pre-war baseline to war period, characterised by a decrease in joy and increases in sadness, fear, and anger.

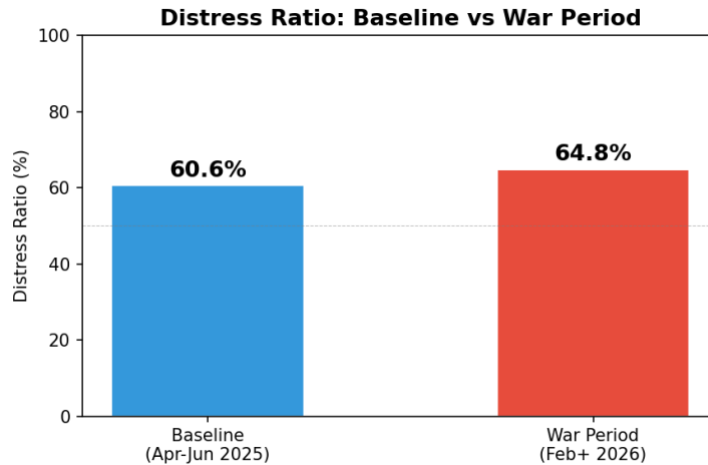


Figure 2. Distress ratio for pre-war baseline vs war period (fine-tuned model). The +4.2pp increase is statistically significant ($p = 1.72e-06$); this reduces to +0.9pp under cross-model validation (Section 4.3).

4.3 Cross-Model Validation

To assess robustness, results were validated using an independent HuggingFace model with a neutral class. Under this model, the distress increase reduces to +0.9 percentage points and is not statistically significant ($p = 0.069$).

This difference arises because the fine-tuned model lacks a neutral class and is forced to assign an emotion to every comment, leading to an overestimation of distress. The cross-model comparison indicates that while the direction of change remains positive, the effect is small and not statistically robust.

Emotion (HF 7-class)	Baseline %	War %	Δ (pp)
anger	12.7%	13.9%	+1.2
disgust	12.4%	13.6%	+1.1
fear	5.7%	5.3%	-0.4
joy	3.7%	2.5%	-1.1
neutral	54.7%	54.3%	-0.4
sadness	3.2%	2.1%	-1.1
surprise	7.6%	8.3%	+0.7

Table 8. Emotion distribution under the HF model for 2,000 baseline and 2,000 war-period Reddit comments. Neutral remains ~54% in both periods, with small increases in anger and disgust and a slight decrease in sadness.

Distress definition	Baseline	War	Δ (pp)	Significance
Strict (sad+fear+anger)	21.6%	21.3%	-0.2	n.s.
Expanded (+disgust)	34.0%	34.9%	+0.9	n.s.
Chi-square over 7 classes	—	—	—	$\chi^2 = 11.68$, $p = 0.069$

Table 9. Distress shift under the HF model. The estimated change reduces from +4.2pp (fine-tuned model) to near zero (strict) or +0.9pp (expanded) and is not statistically significant at $\alpha = 0.05$.

4.4 Event-Level Analysis

Eight major conflict events were analysed by comparing distress levels before and after each event. While 3 out of 8 events show significant increases at $p < 0.05$, none remain significant after Bonferroni correction, indicating that event-level effects are weak and not statistically robust.

Event	Date	Before	After	Δ pp	p-value	Sig?
Trump 10-day ultimatum	02-20	63.9%	65.0%	+1.1	0.325	NO
Trump State of the Union	02-24	65.9%	60.1%	-5.8	0.988	NO
IAEA weapons-grade finding	02-26	61.7%	63.4%	+1.7	0.200	NO
US-Israel strikes begin	02-28	62.0%	64.5%	+2.5	0.040	YES*
Khamenei killed, retal.	03-01	63.4%	65.2%	+1.8	0.030	YES*
Israel strikes IRIB Tehran	03-03	64.5%	66.2%	+1.6	0.018	YES*
UN SC resolution	03-11	63.8%	64.1%	+0.4	0.327	NO
120+ historic sites damaged	03-27	63.5%	63.9%	+0.3	0.330	NO

Table 10. Event-level Mann–Whitney U test results. Three of eight events are significant at $\alpha = 0.05$ (marked *), but none remain significant after Bonferroni correction (adjusted $\alpha = 0.0063$).

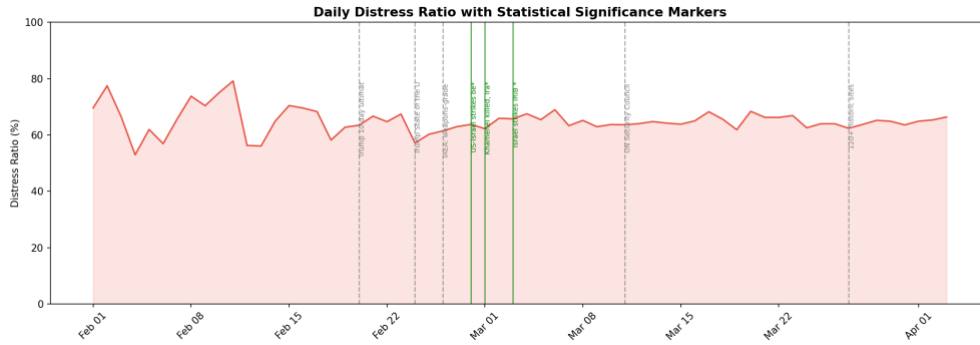


Figure 3. Daily distress ratio with eight conflict events marked. Three events are significant at $\alpha = 0.05$, but none remain significant after Bonferroni correction.

4.5 Community-Level Differences

Distress levels vary across subreddits. News and political communities (e.g., r/worldnews, r/geopolitics) show approximately 2 percentage points higher distress than mental-health subreddits. However, this comparison should be interpreted cautiously due to smaller sample sizes in mental-health communities.

Subreddit	Comments	Distress Ratio	Dominant Emotion
r/worldnews	71,368	65.5%	anger
r/iran	1,612	64.7%	anger
r/geopolitics	13,890	64.1%	anger
r/mentalhealth	369	64.0%	anger
r/anxiety	127	61.4%	anger / fear
r/israel	9,512	60.9%	anger

Table 11. Distress ratio by subreddit during the war period, with news subreddits showing higher volume and slightly higher distress levels.

4.6 Summary of Results

Overall, the analysis suggests a directionally positive but modest increase in distress during the war period. While the fine-tuned model reports a larger effect, cross-model validation indicates that the true effect is small and not statistically significant.

5. Discussion

The results show that while the fine-tuned DistilBERT model detects an increase in distress during the war period, this effect is substantially smaller when validated using an independent, neutral-aware model. The single-model estimate suggests a +4.2pp increase, but cross-model validation reduces this to +0.9pp and renders it statistically non-significant. This indicates a weak positive effect that is not statistically robust, rather than a strong or definitive increase in distress.

A key reason for this discrepancy is the absence of a neutral class in the fine-tuned model. Because the model is trained on a 6-class dataset without a neutral option, it is forced to assign an emotional label to all comments. In contrast, the independent HuggingFace model classifies over half of Reddit comments as neutral. This difference leads to a systematic overestimation of emotional intensity, particularly for anger and joy, in the fine-tuned model.

The analysis also highlights a domain gap between the training data (Saravia dataset) and real-world Reddit text. Although the models achieve high performance on the benchmark dataset ($F1 \approx 0.93$), agreement with an independent LLM judge is much lower (34–41%), indicating reduced reliability in real-world settings. Additional inter-model comparisons further show that agreement between models drops significantly on Reddit data, reinforcing the conclusion that model performance degrades outside the training domain.

Event-level analysis suggests that distress responses are not strongly driven by individual conflict events. While some events appear significant at conventional thresholds, none remain significant after correcting for multiple testing. This indicates that distress is more likely diffuse and cumulative, rather than sharply tied to specific events.

At the community level, news subreddits show slightly higher distress than mental-health subreddits. However, this finding should be interpreted cautiously, as mental-health subreddits have much smaller sample sizes. Overall, the most defensible conclusion is that Reddit discourse reflects a modest and weakly positive shift in distress during the war period, rather than a large-scale mental-health crisis.

6. Limitations

Several limitations should be considered when interpreting the results.

First, the lack of a neutral class in the training data introduces systematic bias. The fine-tuned model is unable to represent neutral or factual comments, leading to inflated estimates of emotional content. Although cross-model validation partially corrects for this, the issue remains a structural limitation of the training setup.

Second, there is a clear domain mismatch between the Saravia dataset and Reddit data. The training data consists of short, well-structured text, while Reddit comments are often longer, informal, and include sarcasm, negation, and multi-clause expressions. This mismatch reduces model accuracy in real-world application.

Third, the analysis relies on Reddit as a data source, which is not representative of the general population. Reddit users tend to be younger, more male, and more digitally engaged, limiting the generalisability of the findings.

Fourth, the dataset is heavily imbalanced across subreddits, with r/worldnews contributing the majority of comments. As a result, aggregate results largely reflect the behaviour of this subreddit rather than a balanced view across communities.

Fifth, the use of keyword-based data collection means the sample is filtered by topic rather than capturing the full range of Reddit discourse. This may introduce selection bias and limit broader interpretation.

Finally, the study identifies associations rather than causal effects. The observed differences between pre-war and war periods cannot be attributed solely to the conflict, as other time-varying factors may also influence emotional expression.

7. Conclusion

This study developed a comparative NLP framework to analyse emotional distress in Reddit discourse during the 2026 Iran–Israel–US war. Among the five methods evaluated, the fine-tuned DistilBERT model achieved the best accuracy–cost tradeoff on the Saravia benchmark and was used as the primary model for large-scale inference.

Initial results from the fine-tuned model indicate a statistically significant increase in distress during the war period (+4.2pp). However, cross-model validation using an independent neutral-aware model reduces this effect to +0.9pp and renders it statistically non-significant. This highlights a key limitation of the 6-class model, which lacks a neutral category and tends to overestimate emotional content.

Across analyses, distress signals are found to be directionally positive but modest. Event-level results show no statistically robust spikes after correction for multiple testing, suggesting that emotional responses are not strongly driven by individual events. Community-level comparisons indicate slightly higher distress in news subreddits, though this finding is constrained by sample imbalance.

Overall, the most defensible conclusion is that Reddit discourse reflects a weak positive shift in distress during the war period, rather than a large-scale or statistically robust mental-health effect. This study demonstrates the importance of cross-model validation and careful interpretation when applying NLP models to real-world social media data.

8. Future Work

Several directions can improve and extend this work.

First, future studies should incorporate a neutral-aware training dataset, such as GoEmotions, to address the forced-emotional-label bias observed in this study. This would allow more accurate estimation of emotional distributions in real-world text.

Second, cross-platform validation using data from sources such as Twitter/X or other social media platforms would help assess whether the findings generalise beyond Reddit.

Third, more advanced causal analysis methods, such as interrupted time series or synthetic control approaches, could be applied to better isolate the effect of the conflict from other time-varying factors.

Fourth, expanding the analysis to include multilingual data would improve coverage, particularly for subreddits where multiple languages are used.

Finally, future work could analyse changes at the user level by tracking how individual users' emotional expressions evolve over time, rather than relying solely on aggregate trends.

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